Brac University

Department of Electrical & Electronic Engineering

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Course Number : EEE321L

9

Section :01

Group No :03

Junaid (20/7/26)

Lab Report

Experiment no: 04

Name of the experiment:

No-Load Performances of overhead transmission line

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...ective :

(i) Measurement of the voltages in no-load operation

(ii) Concept of operating capacitance

(iii) Line model with increased operating capacitance

Theoretical Background :

In a tranmission line is one of the most important components of an electrical power system, responsible for transforing ~~ect~~ electrical energy from generating station to consumers. No-load operation occurs when the sending end is energized while the receiving end remain open or disconnected from any load. Even through no load current flows; the distributed capacitance of the li allows a small charging current to circulate.

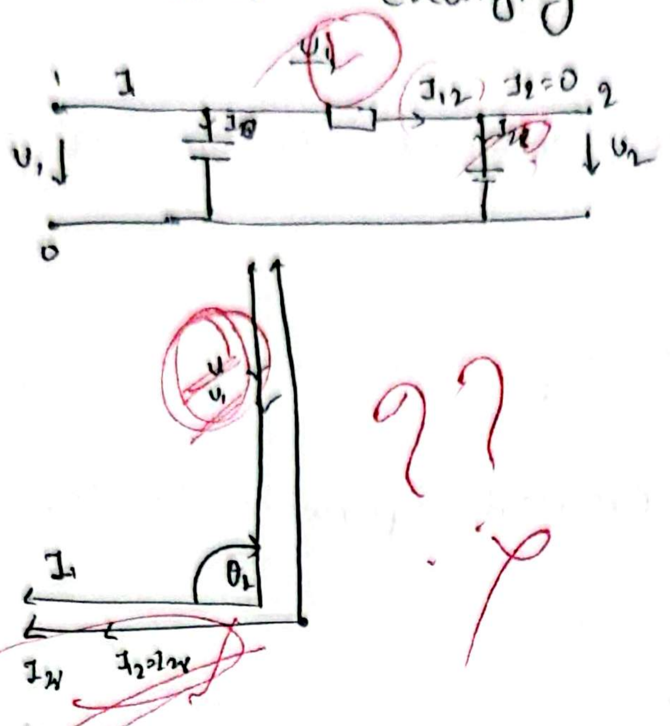


fig: Single-phase Equivalent circuit Diagram of a Loss-free Line in No load operation, with Appropriate vector Diagram.

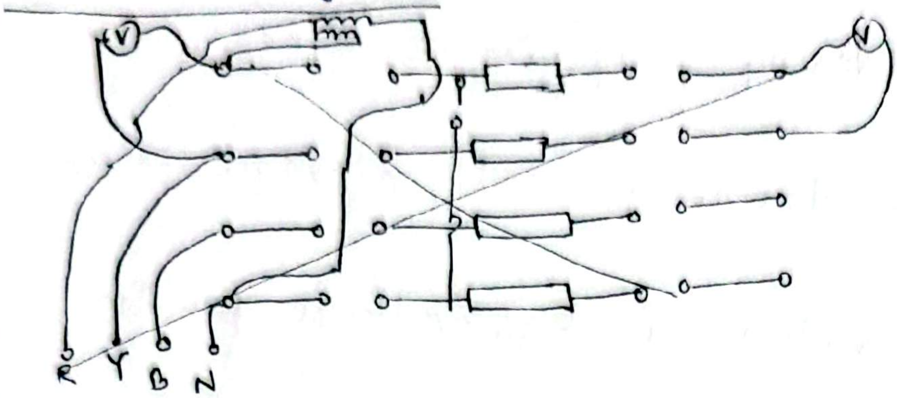
This charging current produces a voltage rise at the receiving end, known as Ferranti effect, which becomes more noticeable as the transmission line length increases. The experiment demonstrates how line capacitance influences voltage distribution no-load condition and highlights the (2)

...portance of considering capacitive effects in transmission system design; voltage regulation and the safe operation of high-voltage power networks.

Apparatus:

1. Three phase supply
2. Overhead Line Model
3. Line Capacitor
4. power Meter
5. Moving - Iron voltmeter

Circuit Diagram:



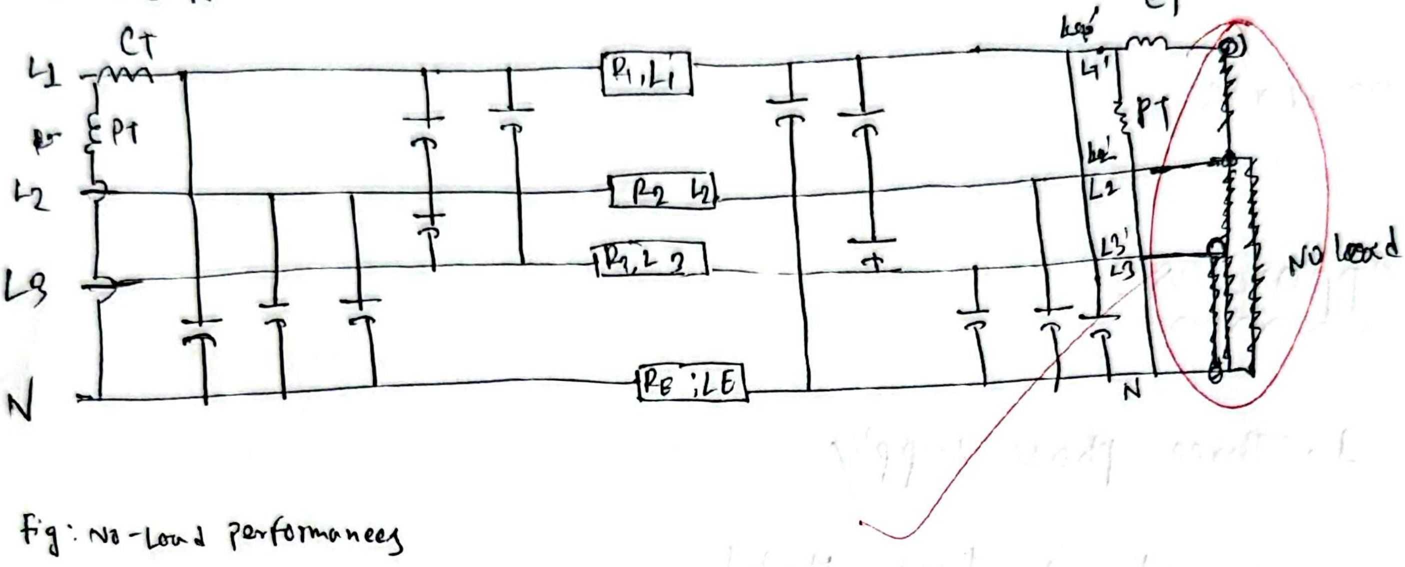


Fig: No-Load performances

Experimental procedure:

Assemble the circuit per the manual (insert bridging plugs so the model capacitances are connected) set the supply to UN = 200 V. and measure the voltages at the beginning and end of the Line and active and reactive powers for one phase. compare the results with those at experiment an equivalent capacitance of the operating capacitance CB performs in the same way as the Individual capacitances CE and CL of the Line.

Hardware Setup:-

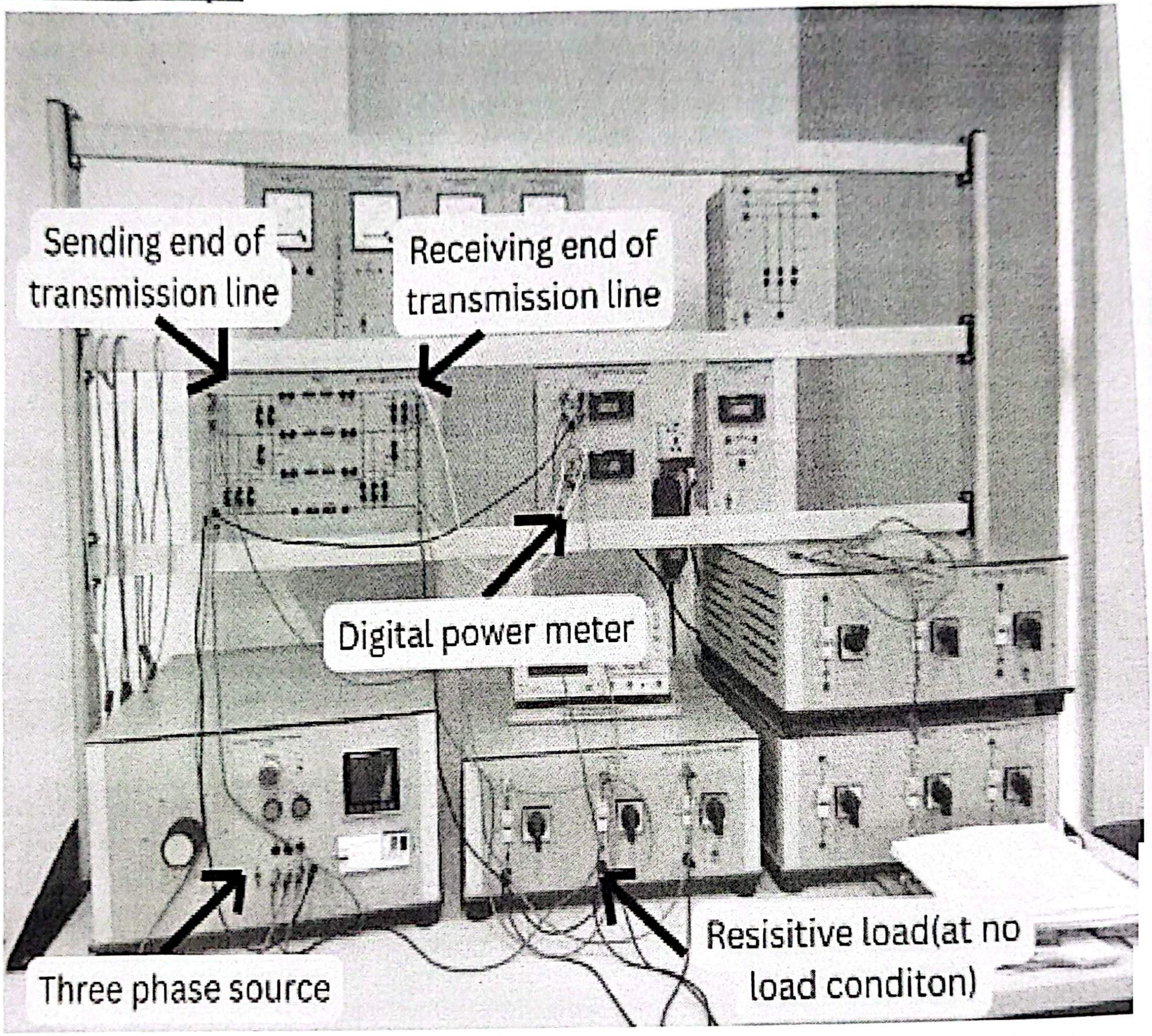


Figure: Hardware setup for no load condition

table :

	I_1	U_1	P_1	U_2	Q_1
All (5μF) Capacitors	0.254	140.8	1.3	157.6	-34
C_E (2μF)	0.108	140.6	0.4	148.4	-12.3
C_L (3μF)	0.163	140.5	0.8	152.3	-20.9
without Capacitors	0	140.3	0	140.3	0

$Q_c = \omega C_B U_N^2 = 2\times\pi\times 50\times 5\times 10^{-6}\times 140.8^2 = 31.1$

$U_1 = 140.8$
$P = 1.3$
$U_2 = 157.6$
$C_B = 5\mu B$
$Q_1 = -34$

Result Analysis :

This experiment result shows that the receiving end voltage increased when capacitors were connected confirming the Ferranti effect. ~~wl~~ without any capacitors, both the sending-end and receiving end voltages were almost equal (140.3) and no charging current flowed.

with the 2μF capacitor, the receiving-end voltage increased to 148.4 V with a charging current of 0.4 A. Using the 3 μF cap... the charging current increased to 0.8 A and the receiving-end voltage rose to 152.3. When all a capacitors (5 μF) were connected, the highest charging current (1.3 A) and the maximum receiving-end voltage (157.6 V) were obtained.

These results indicate that increasing the line capacitance increases the charging current and cause a greater rise in the receiving-end voltage. Therefore, the experiment observations are consistent with the theoretical behavior of a no-load transmission line and successfully verify the Ferranti effect

Question & Answer:

$Q_C = \omega C_B U_N^2 = 31J$ [from data table calculation]

In no load transmission line, load can operate at a very low reactive power

...ssion :

...is experiment successfully demonstrated the behavior of a transmission line under no-load conditions and verified the ferranti effect.

The results showed that the receiving-end voltage increased as the line capacitance increased, while the sending-end voltage remained nearly constant. A higher capacitance also produced a larger charging current, which contributed to the voltage rise at the receiving end. Without any capacitor, both the voltages were almost equal, confirming that the voltage rise was caused by the capacitive effect of the line. The experime observation closely matched the theoreti[cal] expectations, although minor differences may have occurred due to instrument inaccuracies, connection losses, and component tolerances.

Overall, the experiment provided a clear understanding of the influence of line capacitance on transmission line performance under no load conditions.

Experimental Procedure

Assemble the circuit in accordance with the foregoing topographic diagram.

Remove all bridging plugs connecting the capacitances to line model.
Connect the artificial line capacitances to the beginning and to the end of the line model. Set the supply voltage to U_N= 200 V.

Measure the voltage between the outer conductors at the beginning and end of each line capacitance, as well as the reactive power consumed by one of the phases:

U_1 = ... V Q = ... Var(cap) U_2 = ... V

Compare the results with those at experiment 1.1: an equivalent capacitance of the operating capacitance C_B performs in the same way as the individual capacitances C_E and C_L of the line.

Junaid (6/7/26)

Faculty Signature and Date

	I_1	U_1	P_1	U_2	Q_1
All (5μF) Capacitor	0·254	140·8	1·3	157·6	−34
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